159302 – Artificial Intelligence
Tutorial - Propositional Logic

The following notations are used in the questions that follow:

- **^** (caret) for and,
- **v** (lower case v) for or,
- **~** for not,
- **->** (hyphen, >) for implication, and
- **<->** (<, hyphen, >) for biconditionals.
- Propositional variables are upper case letters (and possibly numbers).
- The symbols true and false are written out in lower case.

## I. Precedence

Give a fully parenthesized version of each of these expressions. Each of your answers needs to follow the definitions legal sentences in propositional logic. Preserve the original order of the variable names in your parenthesized expression.

1. P v ~ Q -> R
2. P ^ Q v R
3. P -> Q <-> R -> S
4. ~ R v S -> Q ^ R

## II. Translating Sentences

If we let the propositional variables S, Z, R, and H have the following intended meanings:

- S = the sun shines
- Z = we go to the zoo
- R = it rains
- H = we stay at home

Give, for each of the following English sentences, a propositional logic sentence with the same meaning:

1. If the sun shines then we go to the zoo.
2. It rains or the sun shines.
3. If we do not go to the zoo then we stay at home.
4. If we do not go to the zoo then it rains.
5. If we do not stay at home then the sun shines.
6. If the sun shines or we do not stay at home then we go to the zoo.
7. If the sun does not shine then we do not go to the zoo.

## III. Simplifications

Use the semantics rules to simplify the following formulas as much as possible.

1. P v true
2. P ^ true
3. P v false
4. P ^ false
5. (P ^ false) v Q
6. (P ^ Q) v true
7. P ^ (Q v true)
8. P ^ (Q v false)

*Source: MITOpenCourseware*

## IV. CNF

For each of the sentences below, derive the conjunctive normal form and enter it as a sequence of parenthesized disjunctions separated by and's.

1. (A ^ ~B) -> C
2. (A -> B) -> C
3. A -> (B -> C)
4. ((A ^ B) v C) -> D
5. ((A ^ B) v C) -> (D ^ E)

## V. Propositional Resolution Refutation

Given the following propositional clauses (axioms), derive a contradiction using the resolution rule:

Prove ~P.
1 ~P v Q v S
2 ~S
3 ~Q v X
4 ~X

## VI. Resolution Theorem Proving

**Adams, Brown and Clark**
There are three suspects for a murder: Adams, Brown, and Clark. Adams says "I didn't do it. The victim was an old acquaintance of Brown's. But Clark hated him." Brown states "I didn't do it. I didn't know the guy. Besides I was out of town all week." Clark says "I didn't do it. I saw both Adams and Brown in town around the victim that day; one of them must have done it." We know that exactly one of the suspects is guilty. Assume that the two innocent men are telling the truth, but that the guilty man might not be.

Let the propositional variables have the following definitions:
A = Adams is innocent
B = Brown is innocent
C = Clark is innocent
X = Brown knew the victim
Y = Brown was in town
Z = Adams was out of town
W = Clark liked the victim
P = Brown was with the victim
Q = Adams was with the victim
F = Brown was a friend of the victim
S = Clark knew the victim

The key to the solution is to make the suspect's statements conditional on their innocence. The following rules/axioms are sufficient.

1. If Adams was innocent, then Brown was the victim's friend and Clark did not like the victim, just as Adams said.
2. If Brown was innocent, he wasn't in town and he doesn't know the victim.

3. If Clark was innocent, then Adams was with the victim and Brown was with the victim.
4. General Knowledge:
   If Brown was with the victim, then Brown was in town.
   If Brown was a friend of the victim, the Brown knew the victim.
   If Clark liked the victim, then Clark knew the victim.
5. Now we assert that only one of A, B, and C is the guilty person:
   A v B
   A v C
   B v C